CASCADE Conference 2026

Keysight Device Security Solutions KDS
Introduction to our booth

peter.zuijdervliet@keysight.com

KEYSIGHT

WHO WE ARE

Founded in 2001 as RISCURE, acquired by KEYSIGHT in Febr 2024

We solve challenges related to **security** for **embedded devices** that an attacker can have full access to

Customer problems

- Cryptographic implementation
- Protecting IP
- Protecting Sensitive data
- Authorization and Authentication
- Trojan detection
- Secure Update mechanisms (over-the-air)
- Protection against reverse engineering

How we help them

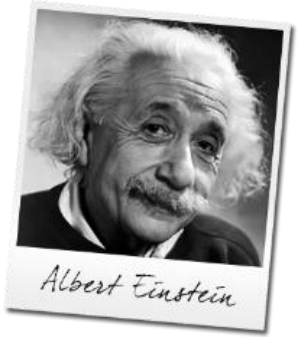
- Simulation test benches
- Physical test benches
- Certification services
- Consultancy / Security testing
- Training courses

How do we help our customers?

Every customer has a different need

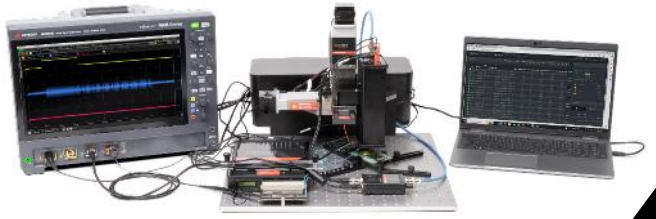
Training

Educate



KEYSIGHT

Riscure Security Solutions



Products

Make

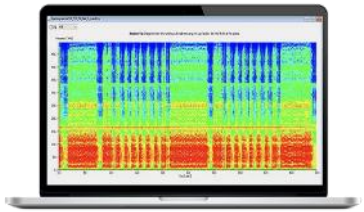
Certification

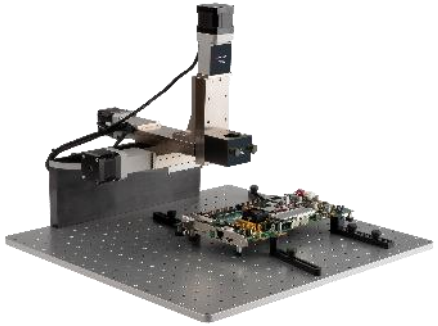
~~Break~~ Test

Products Portfolio

Offering

- +20 specialized devices that can be combined in a solution
- Full coverage of the most advanced security test methods on SCA/FI
- Software subscription for automation and advanced analysis
- Combines with Keysight measurement devices
  - Oscilloscopes
  - Spectrum analyzers
  - Power supplies
  - Digitizers





Automation by Inspector software

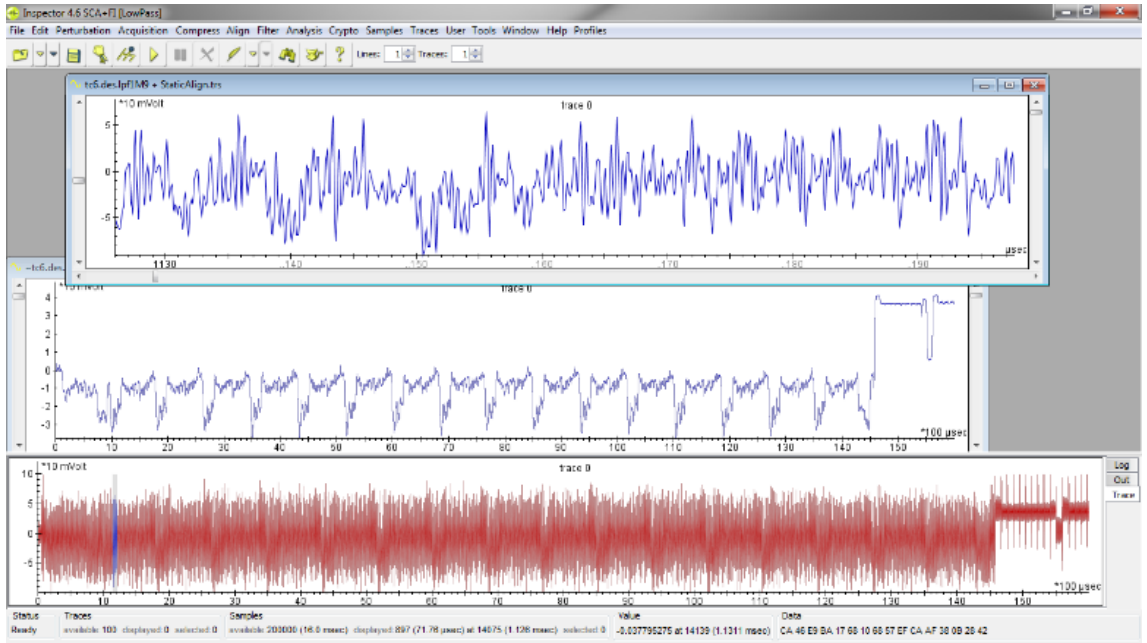
Available in Python and Java, running under Windows and Linux

Side Channel Analysis (SCA)

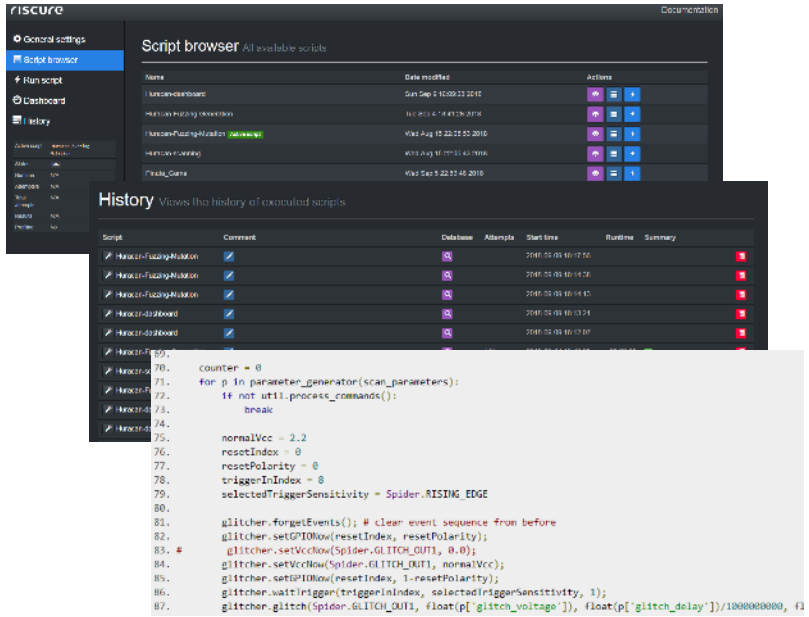
Acquisition
Store/load traces
Compression
Alignment

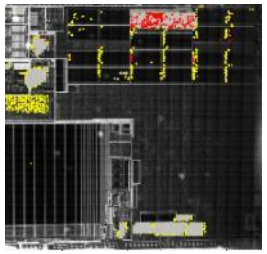
Statistical analysis
Frequency analysis
Filtering
Chaining

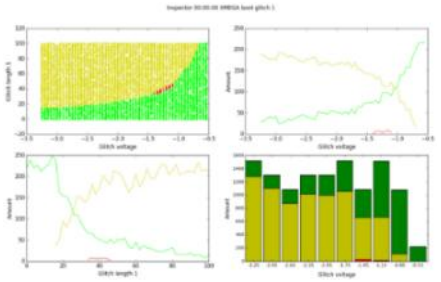
Side channel analysis
Modular design
Functions are modules
IDE to create your own



Fault Injection (FI)







Side Channel Analysis (SCA)

Modular Turnkey solution

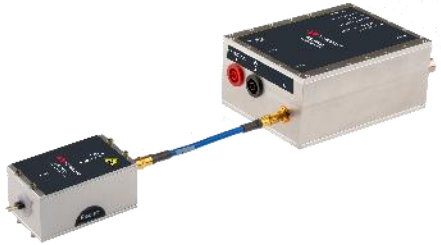
DS1030A / DS1031A
Pinata S / H

MXR-B
Oscilloscope

DS2201A / DS2202A / DS2203A
Inspector Side Channel Analysis

DS1220A
Smart Card
Power Tracer

DS1201A
Passive Current Probe

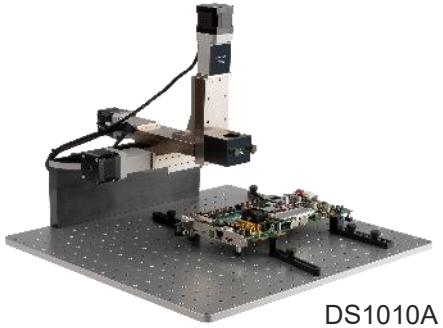


DS1202A
Active Current Probe
With Amplifier Unit (DS1040A)

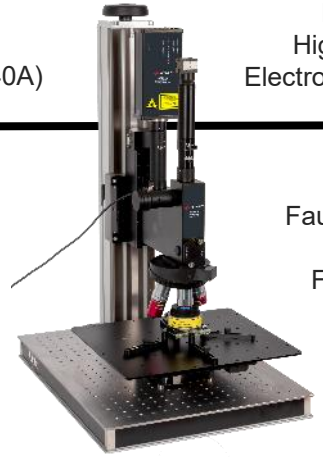
DS1203A
High Precision
Electromagnetic Probe

DS1180A
Glitch Pattern Generator

DS1221A
Notch Filter and Demodulator



DS1010A
Precision XYZ Stage



DS1101A
Fault Injection Laser
System for
Failure Analysis

DS1001A
Transceiver

DS1002A
Pattern Based Trigger
Generator

DS1170A
Automotive Security Test Tool

DS1211A
1425 nm Single Mode Side
Channel Laser Source

DS1210A
1310 nm Single Mode
Side Channel Laser
Source

Fault Injection (FI)

Modular Turnkey Solution

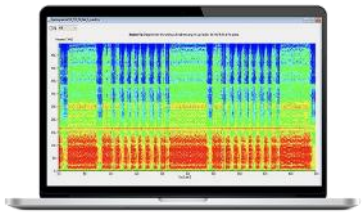
DS1180A
Glitch Pattern Generator

DS1160A
Smartcard Voltage and Clock Glitcher

DS1150A
Clock Glitcher

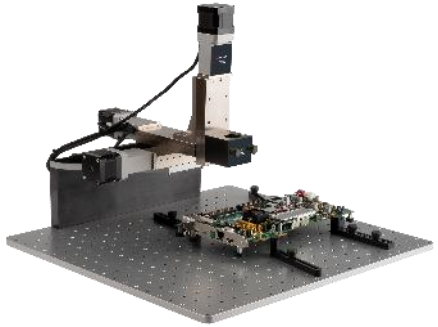
DS1030A / DS1031A
Pinata S / H

MXR-B
Oscilloscope



DS2101A
Inspector Fault Injection

DS1140A
1.5 A Glitch Amplifier



DS1010A
Precision XYZ Stage

DS1121A
Bidirectional Fault Injection Probe

DS1101A
Fault Injection Laser System

DS1102A
Dual Laser Fault Injection System

DS1103A
Compact Laser System

DS1142A
30 A Glitch Amplifier

DS1120A
Unidirectional Fault Injection Probe

DS1110A / DS1111A / DS1112A
445 / 808 / 1064 nm Multimode Fault Injection Laser

DS1001A
Transceiver

DS1002A
Pattern Based Trigger Generator

DS1170A
Automotive Security Test Tool

DS1113A / DS1114A
532 / 1064 nm Diode Pumped Solid State Fault Injection Laser

DS1106A
Safety Box

KEYSIGHT

KEYSIGHT

PXI Embedded Security Testbench

DS1050A

Challenge

What problem do we solve

- **Chips are getting more advanced**
  - **Faster**, meaning higher clock frequencies
  - **Smaller footprint**, 3nm technology
  - Massive, **billions of transistors** on one chip
  - **Richer**, in features and complexity

- **Increase in threats**
  - **Hackers** are also getting **more advanced**; their methods and capabilities get better every year.
  - **Chips are in the field for many years**; how can we test these chips now for the future.

PXIe Migration

One system to make it simpler

DS1050A-Core

- The Inspector software DS2101A, DS2201A, DS2202A and DS2203A will run on the PXIe Chassis and Embedded Controller
- This will create a standalone next generation SCA/FI testbench

DS1050A-001

- The DS1060A is the first product released by Keysight Device Security Testing (former Riscure) within the PXIe platform
- It supersedes the DS1002A – Pattern Based Trigger Generator AKA icWaves and the DS1001A – Transceiver

Current development

- Soon we will release a new PXIe version of the DS1180A Glitch Pattern Generator to upgrade the testbench to a complete replacement of the current testbench

M9046A
Chassis

M9038A
Embedded Controller

Core

Workstation

DS1002A
Pattern Based Trigger Generator

DS1060A
Pattern Based Trigger Generator (PXI)

Debut

DS1001A
Transceiver

DS1070A / DS1071A
New Glitch Pattern Generator

Current
Development

DS1180A
Glitch Pattern Generator

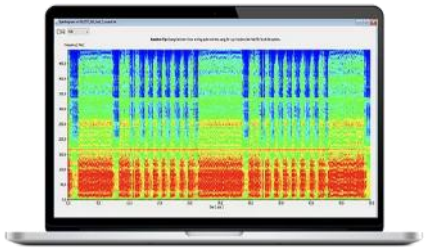
PXIe example setup

Scalable / Modular / Flexible

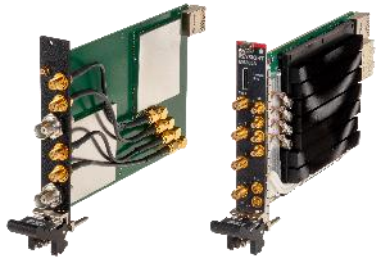
M9046A
Chassis

M9038A
Embedded Controller

DS2201A / DS2202A / DS2203A
Inspector Side Channel Analysis



DS2101A
Inspector Fault Injection



DS1060A
Pattern Based Trigger Generator

M9243A
Oscilloscope
(not recommended)

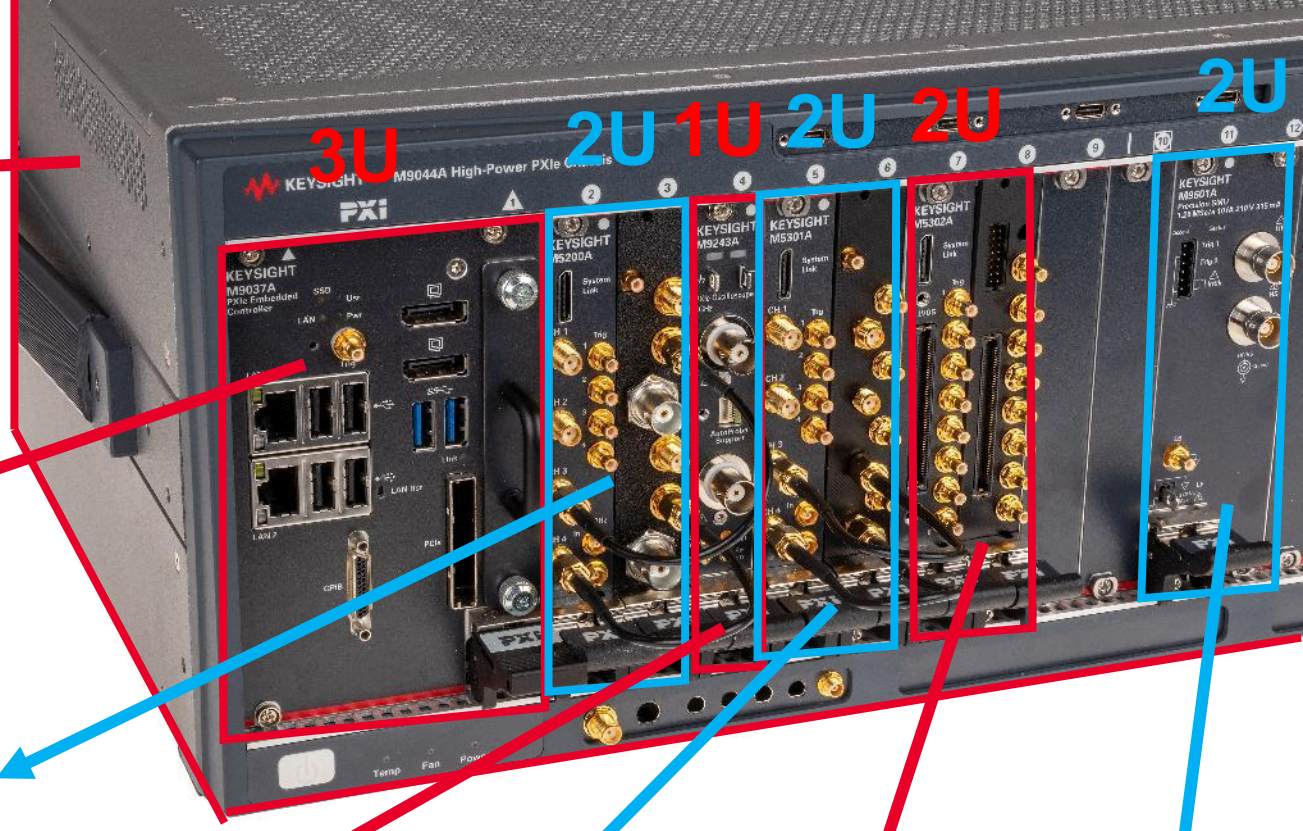
Under Development

DS1070A
AWG + Interface Card

Under Development

DS1071A
Digital IO + Interface Card

M9601A
SMU



Your Next-Gen Device Security Test Lab

Inspector Side Channel Analysis & Fault Injection

Next–Generation Embedded Security Testbench

Clock and Power Solution

Electromagnetic Solution

Laser Solution

Inspector Software

Pre-Silicon

- TVLA
- Leakage Assessment
- Root Cause Analysis

Post-Silicon

- Acquisition
- Signal Processing
- Alignment
- Cryptanalysis

- Communication
- Perturbation
- Differential Fault analysis
- Advance Data Analysis

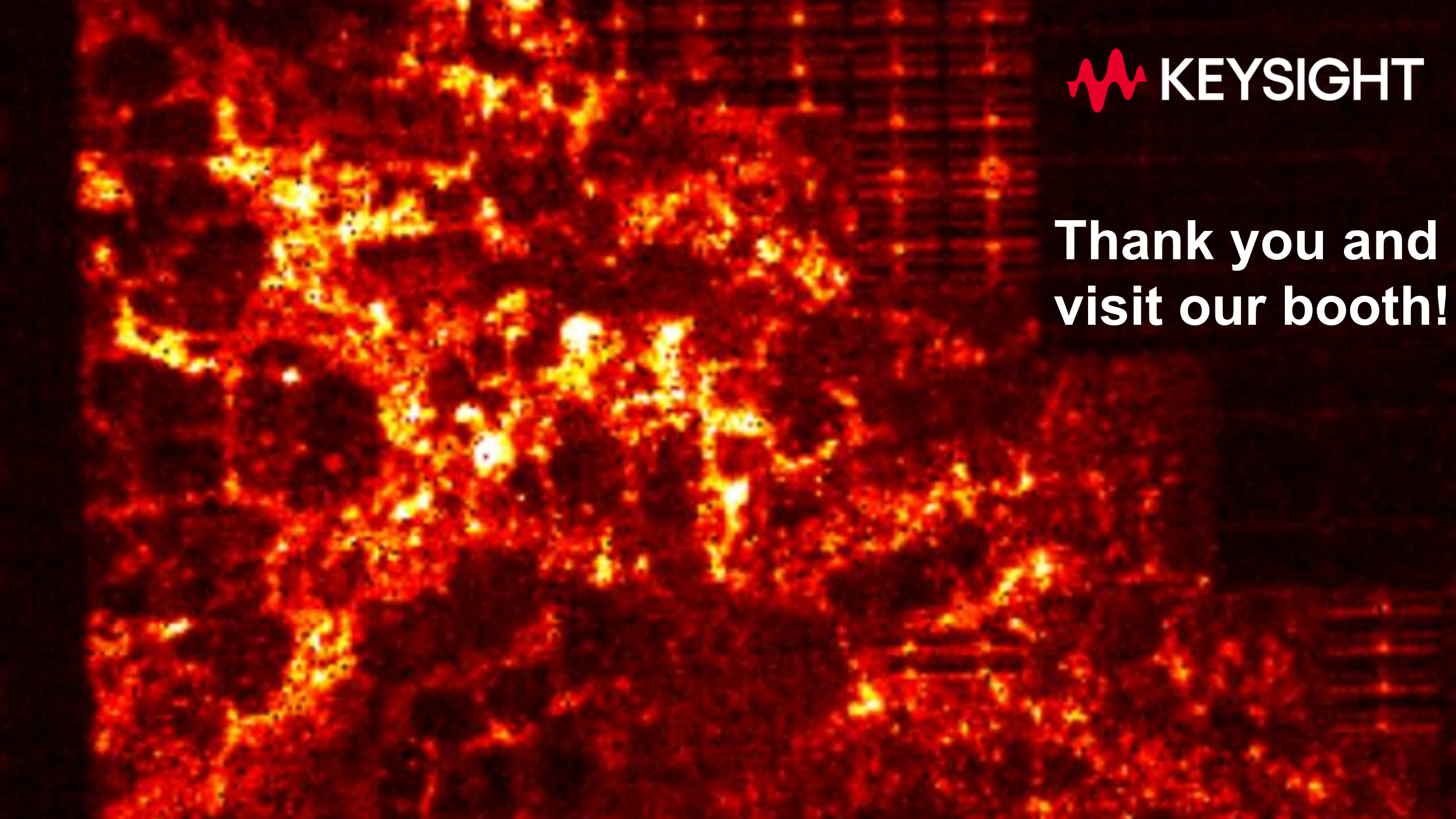
Available in Java

- ✓ 100+ Modules
- ✓ Deep Learning
- ✓ Automation & PQC
- ✓ Headless Mode Support
- ✓ 20 years of real-world use

Side Channel Analysis Training
Essential, Advanced & AI

Fault Injection Training
Essential & Advance

KEYSIGHT



KEYSIGHT

Thank you and visit our booth!